1 Abstract Integration

1.1 The Concept of Measurbility

Definition 1.1.1

1. A collection $\mathfrak{M}$ of subsets of a set X is said to be a ***σ-algebra*** in X if $\mathfrak{M}$ has the following properties:
   (a) $X \in \mathfrak{M}$.
   (b) If $A \in \mathfrak{M}$, then $A^c \in \mathfrak{M}$.
   (c) If $A = \bigcup_{n=1}^{\infty} A_n$ and $A_n \in \mathfrak{M}$ for $n = 1, 2, 3, \cdots$, then $A \in \mathfrak{M}$.
2. If $\mathfrak{M}$ is a σ-algebra in X, then X is called a ***measurable*** provided that $f^{-1}(V)$ is measurable set in X for every open set V in Y.
3. If X is a measurable space, Y is a topological space, and f is a mapping of X into Y, then f is said to be ***measurable*** provided that $f^{-1}(V)$ is a measurable set in X for every open set V in Y.

Note.
对比 topological space 的定义, σ-algebra 的定义是对称的, 即可测集的补集仍是可测的, 而对于拓扑空间, 开集的补集不是开集, 而是被定义为了闭集这样的对象.

Lemma 1.1.2

1. Let X be a measurable space , Y, Z be topological spaces. If $f : X \to Y$ is measurable , and if $g : Y \to Z$ is continuous, then $g \circ f : X \to Z$ is measurable.
2. Let u, v be real measurable functions on a measurable space X, let Φ be a continuous mapping of the plane into a topological space Y, and define
$$h(x) = \Phi(u(x), v(x))$$
for $x \in X$. Then $h : X \to Y$ is measurable.

Corollary 1.1.3

Let X be a measurable space. The following propositions hold

1. If $f = u + iv$, where u and v are real measurable functions on X, then f is a complex measurable function on X.
2. If $f = u + iv$ is a complex measurable function on X, then u, v, and $|f|$ are real measurable functions on X.
3. If f and g are complex measurable functions on X, then so are $f + g$ and fg.
4. If E is a measurable set in X and if

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$$

   then χ_E is a measurable function.
5. If f is a complex measurable function on X , there is a complex measurable function α on X such that $|\alpha| = 1$ and $f = \alpha |f|$.

Theorem 1.1.4

If $\mathscr{F}$ is any collection of subsets of X , there exists a smallest σ-algebra $\mathfrak{M}^*$ such that $\mathscr{F} \subseteq \mathfrak{M}^*$.

Definition 1.1.5 ▸ Borel

Let X be a topological space.

1. By ***Borel σ-algebra*** , we mean the smallest σ-algebra $\mathscr{B}$ in X such that every open set in X belongs to $\mathscr{B}$. The members of $\mathscr{B}$ are called the ***Borel sets*** of X.
2. All countable unions of closed sets and all countable intersections of open sets are Borel sets, which we called F_σ***'s and*** G_δ***'s***, respectively.
3. By a ***Borel function***, we mean a measurable function on the measurable space $(X, \mathscr{B})$.

Remark.

1. The letters F and G were used for colsed and opensets,respectively, and σ refers to union, δ to intersection.
2. A continuous function is Borel-measurable, since the preimage of any open set is open and therefore a Borel set.

Theorem 1.1.6

Suppose $\mathfrak{M}$ is a σ-algbera in X ,and Y is a topological space. Let f map X into Y.

1. If Ω is the collection of all sets $E \subseteq Y$ such that $f^{-1}(E) \in \mathfrak{M}$, then Ω is a σ- algebra in Y.
2. If f is measurable and E is a Borel set in Y , then $f^{-1}(E) \subseteq \mathfrak{M}$.
3. If $Y = [-\infty, \infty]$ and $f^{-1}((\alpha, \infty]) \subseteq \mathfrak{M}$ for every real α, then f is measurable .
4. If f is measurable, if Z is topological space, if $g : Y \to Z$ is a Borel mapping, and if $h = g \circ f$, then $h : X \to Z$ is measurable.

Theorem 1.1.7

If $f_n : X \to [-\infty, \infty]$ is measurable,for $\neq 1, 2, 3, \cdots$, and

$$g = \sup_{n \geq 1} f_n, \quad h = \lim_{n \to \infty} \sup f_n,$$

then g and h are measurable.

Corollary 1.1.8

1. The limit of every pointwise convergent sequence of complex measurable functions is measurable.
2. If f and g are measurable(with range in $[-\infty, \infty]$), then so are $\max\{f, g\}$ and $\min\{f, g\}$. In particular , this is true of the functions

$$f^+ = \max\{f, 0\}, \quad f^- = -\min\{f, 0\}.$$

which are called the ***positive part*** and ***negative part*** of f , respectively.

Remark.

1. There are the standrd representation

$$|f| = f^+ + f^-, \quad f = f^+ - f^-$$

2. And easy but (may) useful observation is : If $f = g - h, g \geq 0, h \geq 0$, then $f^+ \leq g$ and $f^- \leq h$.

1.2 Simple Functions

Definition 1.2.1

A complex function s on a measurable space X whose range consists of only finitely many points will be called a ***simple function***. Among these are the nonnegative simple functions, whose range is a finite subset of $[0, \infty)$.
Specifically, if $\alpha_1, \cdots, \alpha_n$ are distinct values of a simple function s, and if we set $A_i = \{x : s(x) = \alpha_i\}$, then clearly

$$s = \sum_{i=1}^{n} \alpha_i \chi_{A_i}.$$

Where χ_{A_i} is the characteristic function of A_i.

Remark.
- Here, we explicity exclude ∞ from the values of a simple function.
- It is clear that s is mearuable if and only if each of the sets A_i is measurable.

Theorem 1.2.2

Let $f : X \to [0, \infty]$ be measurable space. There exists simple measurable functions s_n on X such that
1. $0 \leq s_1 \leq s_2 \leq \cdots \leq f$.
2. $s_n(x) \to f(x)$ as $n \to \infty$, for every $x \in X$.

Proof sketch.
We construst a sequence of Borel simple functions $\{\varphi_n(x)\}$ to act as an ***identify***. 当 n 增大的同时，我们同时让 $\varphi_n(x)$ 的单位逼近精度和单位逼近范围随着 n 提升. 并且在舍弃误差时，总是向下取整，使得该单位逼近是自下而上的.

Proof. For every $x \in [0, \infty]$, and for every $n \in \mathbb{N}$, there exists a unique integer $k_n(x)$, such that

$$k_n(x) 2^{-n} \leq x < (k_n(x) + 1) 2^{-n}$$

For every $n \in \mathbb{N}$, we define

$$\varphi_n(x) = \begin{cases} k_n(x)\,2^{-n}, & 0 \leq x \leq n \\ n, & x \geq n \end{cases}$$

Each $\varphi_n(x)$ is then a Borel simple function. It is not hard to show that $0 \leq \varphi_1 \leq \varphi_2 \cdots \leq \cdots \leq \mathrm{Id}$. For each n, we define

$$s_n(x) := (\varphi_n \circ f)(x)$$

Then $\{s_n\}$ is a suquence of simple measurable functions such that $0 \leq s_1 \leq s_2 \leq \cdots \leq f$. Since $\lim_{n\to\infty} \varphi_n(x) = x$, then $\lim_{n\to\infty} s_n(x) = f(x)$. □

1.3 Measure

Definition 1.3.1 ▸ Measure and Measure Space

(a) A ***positive measure*** is a function μ, defined on a σ-algebra $\mathfrak{M}$, whose range is in $[0, \infty]$ and which is ***countably additive***. This means that if $\{A_i\}$ is a disjoint countable collection of members of $\mathfrak{M}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

To avoid trivialities, we shall also assume that $\mu(A) < \infty$ for at least one $A \in \mathfrak{M}$.

(b) A ***measure space*** is a measurable space which has a positive measure defined on the σ-algebra of its measurable sets.

(c) A ***complex measure*** is a complex-valued countably additive function defined on a σ-algebra.

Theorem 1.3.2

Let μ be a positive measure on a σ-algebra $\mathfrak{M}$. Then

1. $\mu(\varnothing) = 0$.
2. $\mu(A_1 \cup \cdots \cup A_n) = \mu(A_1) + \cdots + \mu(A_n)$ if $A_1, \ldots, A_n$ are pairwise disjoint members of $\mathfrak{M}$.
3. $A \subset B$ implies $\mu(A) \leq \mu(B)$ if $A \in \mathfrak{M}, B \in \mathfrak{M}$.

4. $\mu(A_n) \to \mu(A)$ as $n \to \infty$ if $A = \bigcup_{n=1}^{\infty} A_n, A_n \in \mathfrak{M}$, and

$$A_1 \subset A_2 \subset A_3 \subset \dots$$

5. $\mu(A_n) \to \mu(A)$ as $n \to \infty$ if $A = \bigcap_{n=1}^{\infty} A_n, A_n \in \mathfrak{M}$,

$$A_1 \supset A_2 \supset A_3 \supset \dots,$$

and $\mu(A_1)$ is finite.

Proof. 1. Take $A \in \mathfrak{M}$ such that $\mu(A) < \infty$. [1] And let $A_2 = A_3 = \cdots = \varnothing$, then $\mu(\varnothing) > 0$ leads to a contradition to the countably additive.

2. Take $A_{n+1} = A_{n+2} = \cdots = \varnothing$.

3. Note that $B = (B \setminus A) \cup A$, then by additivity

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$$

4. Let $A_0 = \varnothing$, and let $B_n = A_n \setminus A_{n-1}$ for all $n \in \mathbb{N}$. Then $B_1, \cdots, B_n$ are pairwise disjoint members of $\mathfrak{M}$ such that $A = \bigcup_{n=1}^{\infty} B_n$. We have

$$\mu(A) = \sum_{n=1}^{\infty} \mu(B_n) = \sum_{n=1}^{\infty} \mu(A_n \setminus A_{n-1})$$

If one of the $\mu(A_n)$ is ∞, then $\lim_{n\to\infty} \mu(A_n)$ and $\mu(A)$ both are ∞ . Otherwise, we have

$$\mu(A_n \setminus A_{n-1}) = \mu(A_n) - \mu(A_{n-1}), \quad \forall n \mathbb{N}$$

Thus

$$\mu(A) = \sum_{n=1}^{\infty} \mu(A_n \setminus A_{n-1}) = \lim_{n\to\infty} \mu(A_n)$$

5. Let $B_n = A_n \setminus A_{n+1}$ for all $n \in \mathbb{N}$. $B_1, \cdots, B_n$ are pariwise disjoint members of $\mathfrak{M}$ with finite measure, such that

$$A_1 \setminus A = A_1 \setminus \left(\bigcap_{n=1}^{\infty} A_n\right) = \bigcup_{n=1}^{\infty} (A_1 \setminus A_n) = \bigcup_{n=1}^{\infty} \left(\bigcup_{k=1}^{n} B_k\right) = \bigcup_{n=1}^{\infty} B_n$$

[1] That is what we supposed at the definition of measure.

. We have

$$\mu(A_1 \setminus A) = \mu\left(\bigcup_{n=1}^{\infty} B_n\right),$$

where the RHS is $\mu(A_1) - \mu(A)$, and the LHS is

$$\sum_{n=1}^{\infty} (\mu(A_n) - \mu(A_{n+1})) = \mu(A_1) - \lim_{n\to\infty} \mu(A_{n+1})$$

Since $\mu(A_1) < \infty$, we have

$$\mu(A) = \lim_{n\to\infty} \mu(A_n)$$

$\square$

Example 1.3.3

1. For any $E \subset X$, where X is any set, define $\mu(E) = \infty$ if E is an infinite set, and let $\mu(E)$ be the number of points in E if E is finite. This μ is called the ***counting measure*** on X.
2. Fix $x_0 \in X$, define $\mu(E) = 1$ if $x_0 \in E$ and $\mu(E) = 0$ if $x_0 \notin E$, for any $E \subset X$. This μ may be called the ***unit mass concentrated at*** x_0.
3. Let μ be the counting measure on the set $\{1, 2, 3, \dots\}$, let $A_n = \{n, n+1, n+2, \dots\}$. Then $\bigcap A_n = \varnothing$ but $\mu(A_n) = \infty$ for $n = 1, 2, 3, \dots$. This shows that the hypothesis

$$\mu(A_1) < \infty$$

is not superfluous in Theorem 1.3.2(5).

1.4 Integration of Positive Funtions

Definition 1.4.1 ▶ Integration of Positive Funtions

1. If $s : X \to [0, \infty)$ is a measurable simple function, of the form

$$s = \sum_{i=1}^{n} \alpha_i \chi_{A_i},$$

where $\alpha_1, \dots, \alpha_n$ are the distinct values of s, and if $E \in \mathfrak{M}$, we define

$$\int_E s\, d\mu = \sum_{i=1}^{n} \alpha_i \mu(A_i \cap E).$$

The convention $0 \cdot \infty = 0$ is used here; it may happen that $\alpha_i = 0$ for some i and that $\mu(A_i \cap E) = \infty$.

2. If $f : X \to [0, \infty]$ is measurable, and $E \in \mathfrak{M}$, we define

$$\int_E f\, d\mu = \sup \int_E s\, d\mu,$$

the supremum being taken over all simple measurable functions s such that $0 \le s \le f$. The left member above is called the ***Lebesgue integral*** of f over E, with respect to the measure μ. It is a number in $[0, \infty]$.

Remark.
We apparently have two definitions for $\int_E f\, \mathrm{d}\mu$ if f is simple, they are the same.

Corollary 1.4.2

(a) If $0 \le f \le g$, then $\int_E f\, d\mu \le \int_E g\, d\mu$.
(b) If $A \subset B$ and $f \ge 0$, then $\int_A f\, d\mu \le \int_B f\, d\mu$.
(c) If $f \ge 0$ and c is a constant, $0 \le c < \infty$, then

$$\int_E cf\, d\mu = c \int_E f\, d\mu.$$

(d) If $f(x) = 0$ for all $x \in E$, then $\int_E f\, d\mu = 0$, even if $\mu(E) = \infty$.
(e) If $\mu(E) = 0$, then $\int_E f\, d\mu = 0$, even if $f(x) = \infty$ for every $x \in E$.
(f) If $f \ge 0$, then $\int_E f\, d\mu = \int_X \chi_E f\, d\mu$.

Remark.
We can also regard the $\int_E f\, \mathrm{d}\mu$ as the restricted function $f|_E$ integrates on the induced measure space E.

Proposition 1.4.3

Let s and t be nonnegative measurable simple functions on X. For $E \in \mathfrak{M}$, define

$$\varphi(E) = \int_E s \, d\mu.$$

Then φ is a measure on $\mathfrak{M}$. Also

$$\int_X (s+t) \, d\mu = \int_X s \, d\mu + \int_X t \, d\mu.$$

Note.
$\varphi(E) = \int_E s \, \mathrm{d}\mu$ 就是依据 s 对测度 μ 进行一个有限的算数的加权. 自然地也就成为一个测度.

Proof. If $E_1, E_2, \cdots$ are disjoint members of $\mathfrak{M}$ whose union is E, the countable additivity of μ shows that

$$\begin{aligned}
\varphi(E) &= \sum_{i=1}^{n} \alpha_i \mu(A_i \cap E) \\
&= \sum_{i=1}^{n} \sum_{j=0}^{\infty} \alpha_i \mu(A_i \cap E_j) \\
&= \sum_{j=0}^{\infty} \sum_{i=1}^{n} \alpha_i \mu(A_i \cap E_j) \\
&= \sum_{j=0}^{\infty} \varphi(E_j)
\end{aligned}$$

Also, $\varphi(\varnothing) \neq \infty$, μ is not identically ∞.

Next, let $\beta_1, \cdots, \beta_m$ be distinct values of t. And let $B_j := \{x : t(x) = \beta_j\}$.

Then

$$\begin{aligned}
\int_X (s+t)\,\mathrm{d}\mu &= \sum_{i,j} \int_{A_i \cap B_j} (s+t)\,\mathrm{d}\mu \\
&= \sum_{i,j} \alpha_i \mu\left(A_i \cap B_j\right) + \sum_{i,j} \beta_j \mu\left(A_i \cap B_j\right) \\
&= \sum_i \alpha_i \mu\left(A_i\right) + \sum_j \beta_j \mu\left(B_j\right) \\
&= \int_X s\,\mathrm{d}\mu + \int_X t\,\mathrm{d}\mu
\end{aligned}$$

□

Theorem 1.4.4 ▸ Lebesgue's Monotone Convergence Theorem

Let $\{f_n\}$ be a sequence of measurable functions on X, and suppose that
(a) $0 \le f_1(x) \le f_2(x) \le \cdots \le \infty$ for every $x \in X$,
(b) $f_n(x) \to f(x)$ as $n \to \infty$, for every $x \in X$.
Then f is measurable, and

$$\int_X f_n\, d\mu \to \int_X f\, d\mu \quad \text{as } n \to \infty.$$

Proof. Since $\int_X f_n\,\mathrm{d}\mu \le \int_X f_{n+1}\,\mathrm{d}\mu$ for all n, there exists an $\alpha > 0$ such that

$$\int_X f_n\,\mathrm{d}\mu \to \alpha, \quad \text{as } n \to \infty$$

By Theorem 1.1.7, f is measurable, then by $\int_X f_n\,\mathrm{d}\mu \le \int_X f\,\mathrm{d}\mu\ \forall n$,

$$\alpha \le \int_X f\,\mathrm{d}\mu$$

Take any simple function $0 \le s \le f$, and $0 \le c < 1$. Define

$$E_n^s := \{x \in X : f_n(x) \ge cs(x)\}$$

Then since $\varphi(E) := \int_E s\,\mathrm{d}\mu$ is a measure ,

$$\alpha \ge \lim_{n\to\infty} \int_X f_n\,\mathrm{d}\mu \ge \lim_{n\to\infty} \int_{E_n^s} f_n\,\mathrm{d}\mu \ge c \lim_{n\to\infty} \int_{E_n^s} s\,\mathrm{d}\mu = c \lim_{n\to\infty} \varphi(E_n^s) = c\varphi(X)$$

That is

$$\alpha \geq c \int_X s \, \mathrm{d}\mu$$

Since s is arbitrary,

$$\alpha \geq c \int_X f \, \mathrm{d}\mu$$

and since c is arbitrary ,

$$\alpha \geq \int_X f \, \mathrm{d}\mu$$

which completes the proof. $\square$

Corollary 1.4.5

Let $f, g : X \to [0, \infty]$ be mearuable functions on X . Then $f + g$ is mearuable and

$$\int_X (f+g) \, \mathrm{d}\mu = \int_X f \, \mathrm{d}\mu + \int_X g \, \mathrm{d}\mu$$

Proof. Let $\{s_n\}, \{s'_n\}$ be simple function sequences in Theorem 1.2.2 of f, g , respectively . Then by the theorem above and the Proposition 1.4.3

$$\begin{aligned}\int_X (f+g) \, \mathrm{d}\mu = \lim_{n\to\infty} \int_X (s_n + s'_n) \, \mathrm{d}\mu &= \lim_{n\to\infty} \int_X s_n \, \mathrm{d}\mu + \lim_{n\to\infty} \int_X s'_n \, \mathrm{d}\mu \\ &= \int_X f \, \mathrm{d}\mu + \int_g \mathrm{d}\mu\end{aligned}$$

$\square$

Corollary 1.4.6

If $f_n : X \to [0, \infty]$ is ***measurable***, for $n = 1, 2, 3, \ldots$, and

$$f(x) = \sum_{n=1}^{\infty} f_n(x) \quad (x \in X),$$

then

$$\int_X f \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu.$$

Proof. Let $g_n(x) = \sum_{k=1}^{n} f_k(x)$. Then $g_1 \leq g_2 \leq \cdots \leq \infty$ and $f = \lim_{n\to\infty} g_n(x)$. Applying Monotone Convergence Theorem to $\{g_n\}$ and f , we have

$$\int_X f \, \mathrm{d}\mu = \lim_{n\to\infty} \int_X \sum_{k=1}^{n} f_k(x) \, \mathrm{d}\mu = \lim_{n\to\infty} \sum_{k=1}^{n} \int_X f_k(x) \, \mathrm{d}\mu = \sum_{n=1}^{\infty} \int_X f_n \, \mathrm{d}\mu$$

□

Corollary 1.4.7

If $a_{ij} \geq 0$ for i and $j = 1, 2, 3, \ldots$, then

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}.$$

Proof. Let μ be counting measure on $\mathbb{Z}$. Let

$$f_i(j) = a_{ij}, \quad f(k) = \sum_{i=1}^{\infty} f_i(k) = \sum_{i=1}^{\infty} a_{ik}$$

By Monotone Convergence Theorem , for any measurable g ,

$$\int_{\mathbb{Z}} g \, \mathrm{d}\mu = \int_{\mathbb{Z}} \lim_{n\to\infty} \chi(\{1, \cdots, n\}) \, g \, \mathrm{d}\mu = \lim_{n\to\infty} \int_{\mathbb{Z}} \chi(\{1, \cdots, n\}) \, g \, \mathrm{d}\mu = \sum_{k=\infty}^{n} g(k)$$

Thus

$$\int_{\mathbb{Z}} f \, \mathrm{d}\mu = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}$$

And we have

$$\sum_{i=1}^{\infty} \int_X f_i \, \mathrm{d}\mu = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij}$$

The above corollary shows that

$$\int_{\mathbb{Z}} f \, \mathrm{d}\mu = \sum_{i=1}^{\infty} \int_X f_i \, \mathrm{d}\mu$$

That is

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}$$

□

Lemma 1.4.8 ▸ Fatou's Lemma

If $f_n : X \to [0, \infty]$ is ***measurable***, for each positive integer n, then

$$\int_X \left(\lim_{n\to\infty} \inf f_n \right) d\mu \leq \lim_{n\to\infty} \inf \int_X f_n d\mu. \tag{1}$$

Note.
有了 Monotone Convergence Theorem 之后, 关键的不等式就剩下

$$\int_X \inf_{n\geq k} f_n \, \mathrm{d}\mu \leq \inf_{n\geq k} \int_X f_n \, \mathrm{d}\mu$$

它的含义是局部最小的积分小于等于积分的最小.

Proof. Let

$$g_k(x) = \inf_{n\geq k} f_n(x)$$

Then

$$\lim_{n\to\infty} \inf f_n = \lim_{n\to\infty} g_n =: g$$

where $\{g_k\}$, g are all measurable . It is obvious that $\{g_k\}$ is increasing . Thus

$$\lim_{n\to\infty} \int_X g_n \, \mathrm{d}\mu = \int_X g \, \mathrm{d}\mu = \int_X \left(\lim_{n\to\infty} \inf f_n \right) \mathrm{d}\mu$$

Only need to show that

$$\int_X \inf_{n\geq k} f_n \, \mathrm{d}\mu \leq \inf_{n\geq k} \int_X f_n \, \mathrm{d}\mu$$

which is true since

$$\int_X \inf_{n\geq k} f_n \, \mathrm{d}\mu \leq \int_X f_n \, \mathrm{d}\mu, \text{for all } n \geq k$$

□

Theorem 1.4.9

Suppose $f : X \to [0, \infty]$ is measurable, and

$$\varphi(E) = \int_E f \, d\mu \quad (E \in \mathfrak{M}).$$

Then φ is a measure on $\mathfrak{M}$, and

$$\int_X g \, d\varphi = \int_X gf \, d\mu$$

for every measurable g on X with range in $[0, \infty]$.

Note.
将 measurable function f 视为对 measure μ 的某种 countable 次算数加权操作. 那么得益于 μ 对 countable 操作的相容性, φ 也就成为一个测度. φ 的积分行为也验证了这种看法的合理性.

Proof.

- Let $E_1, \cdots, E_n, \cdots$ be disjoint measurable sets such that $E = \bigcup_{n=1}^{\infty} E_n$. Then $\left\{\sum_{k=1}^{n} \chi_{E_k} f\right\}$ be increasing measurable function sequence such that $\lim_{n\to\infty} \sum_{k=1}^{n} \chi_{E_k} f = f$. The Monotone Convergence Theorem shows that

$$\sum_{n=1}^{\infty} \varphi(E_k) = \lim_{n\to\infty} \sum_{k=1}^{n} \int_{E_k} f \, \mathrm{d}\mu = \lim_{n\to\infty} \int_E \sum_{k=1}^{n} \chi_{E_k} f \, \mathrm{d}\mu$$
$$= \int_E f \, \mathrm{d}\mu = \varphi(E)$$

  Since $\varphi(\varnothing) = 0$. φ is a measure on X .

- To show the second results. See that it is true for $g = \chi_E$. Thus it is true for g is a simple function . And the general case follows from the

Monotone Convergence Theorem.

□

1.5 Integration of Complex Functions

Definition 1.5.1 ▸ Lebegsgue Integrability

We define $L^1(\mu)$ to be the collection of all complex measurable functions f on X for which

$$\int_X |f|\, d\mu < \infty.$$

Note that the measurability of f implies that of $|f|$, hence the above integral is defined.
The members of $L^1(\mu)$ are called ***Lebesgue integrable functions*** (with respect to μ) or ***summable functions***.

Definition 1.5.2 ▸ Complex Integration

If $f = u + iv$, where u and v are real measurable functions on X, and if $f \in L^1(\mu)$, we define

$$\int_E f\, d\mu = \int_E u^+\, d\mu - \int_E u^-\, d\mu + i\int_E v^+\, d\mu - i\int_E v^-\, d\mu \tag{1}$$

for every measurable set E.

Remark.
We have $u^+ \le |u| < |f|$, etc., so that each of these four integrals is finite.
Thus (1) defines the integral on the left as a complex number.

Definition 1.5.3 ▸ Real Integration

Occasionally it is desirable to define the integral of a measurable function f with range in $[-\infty, \infty]$ to be

$$\int_E f\, d\mu = \int_E f^+\, d\mu - \int_E f^-\, d\mu, \tag{2}$$

provided that at least one of the integrals on the right of (2) is finite. The left side of (2) is then a number in $[-\infty, \infty]$.

Theorem 1.5.4

Suppose f and $g \in L^1(\mu)$ and α and β are complex numbers. Then $\alpha f + \beta g \in L^1(\mu)$, and

$$\int_X (\alpha f + \beta g)\, d\mu = \alpha \int_X f\, d\mu + \beta \int_X g\, d\mu$$

Proof sketch.
可积性来自于模长的三角不等式. 主要证明可加性. 将 $f + g = h$ 的正负部分到两边, 就都变成了非负函数. 积分即可.

Theorem 1.5.5

If $f \in L^1(\mu)$, then

$$\left|\int_X f\, d\mu\right| \leq \int_X |f| d\mu.$$

Proof. Let $z = \int_X f\, \mathrm{d}\mu$, z is a complex number . There exists a complex number α ($\alpha = \frac{\bar{z}}{|z|}$) , such that $|\alpha| = 1$, and

$$\left|\int_X f\, \mathrm{d}\mu\right| = \alpha z = \alpha \int_X f\, \mathrm{d}\mu = \int_X \alpha f\, \mathrm{d}\mu$$

Suppose $\alpha f = u + iv$, since $\int_X \alpha f\, \mathrm{d}\mu$ is a real number , $\int_X v\, \mathrm{d}\mu = 0$. We have

$$\int_X \alpha f\, \mathrm{d}\mu = \int_X u\, \mathrm{d}\mu \leq \int_X |\alpha f|\, \mathrm{d}\mu = \int_X |f|\, \mathrm{d}\mu$$

□

Theorem 1.5.6 ▸ Lebesgue's Dominated Convergence Theorem

Suppose $\{f_n\}$ is a sequence of complex measurable functions on X such that

$$f(x) = \lim_{n\to\infty} f_n(x)$$

exists for every $x \in X$. If there is a function $g \in L^1(\mu)$ such that

$$|f_n(x)| \leq g(x) \quad (n = 1, 2, 3, \ldots ; x \in X),$$

then $f \in L^1(\mu)$,

$$\lim_{n \to \infty} \int_X |f_n - f| \, d\mu = 0,$$

and

$$\lim_{n \to \infty} \int_X f_n \, d\mu = \int_X f \, d\mu.$$

Note.
$\{f_n\}$ 不一定单调, 能将极限从积分外塞进积分里的工具, 目前我们只有 Fatou's Lemma , 但是它原始的样子跟我们需要的不等号是反向的. 自然的想法是填一个负号, 这又会导致函数失去非负性, 而控制函数 g 恰好能弥补这一点不足. 实现起来, 我们从 g 上扣去 f, f_n , 而非直接拿出 f, f_n 去做.

Proof. Since $|f| \leq g$ and f is measurable , $f \in L^1(\mu)$. Function $(2g - |f_n - f|)$ is a nonnegative measurable function . Applying Fatou's Lemma to it , we have

$$\int_X \liminf_{n \to \infty} (2g - |f_n - f|) \, \mathrm{d}\mu \leq \liminf_{n \to \infty} \int_X (2g - |f_n - f|) \, \mathrm{d}\mu$$

where

$$\liminf_{n \to \infty} (2g - |f_n - f|) = 2g - \limsup_{n \to \infty} |f_n - f| = 2g$$

$$\liminf_{n \to \infty} \int_X (2g - |f_n - f|) \, \mathrm{d}\mu = 2 \int_X g \, \mathrm{d}\mu - \limsup_{n \to \infty} \int_X |f_n - f| \, \mathrm{d}\mu$$

The original inequality comes to

$$2 \int_X g \, \mathrm{d}\mu \leq 2 \int_X g \, \mathrm{d}\mu - \limsup_{n \to \infty} \int_X |f_n - f| \, \mathrm{d}\mu$$

Since $\int_X g \, \mathrm{d}\mu$ is finite, we can substract $2 \int_X g \, \mathrm{d}\mu$ both sides and get

$$\limsup_{n \to \infty} \int_X |f_n - f| \, \mathrm{d}\mu \leq 0$$

Thus

$$\lim_{n\to\infty} \int_X |f_n - f| \, d\mu = 0$$

Finnaly , we have

$$\begin{aligned}\left|\lim_{n\to\infty} \int_X f_n \, d\mu - \int_X f \, d\mu\right| &= \left|\lim_{n\to\infty} \int_X (f_n - f) \, d\mu\right| \\ &\leq \lim_{n\to\infty} \int_X |f_n - f| \, d\mu \\ &= 0\end{aligned}$$

which completes the proof. □

1.6 The Sets of Measure Zero

Definition 1.6.1

Let P be a property which a point x may or may not have. If μ is a measure on a σ-algebra $\mathfrak{M}$ and if $E \in \mathfrak{M}$, the statement "P holds ***almost everywhere on*** E" (abbreviated to "P holds a.e. on E") means that there exists an $N \in \mathfrak{M}$ such that $\mu(N) = 0$, $N \subset E$, and P holds at every point of $E - N$.

Remark.
This concept of a.e. depends of course very strongly on the given measure, and we shall write "a.e. [μ]" whenever clarity requires that the measure be indicated.

Example 1.6.2

For example, if f and g are measurable functions and if

$$\mu(\{x : f(x) \neq g(x)\}) = 0,$$

we say that $f = g$ a.e. [μ] on X, and we may write $f \sim g$.

Remark.
This is easily seen to be an equivalence relation. The transitivity ($f \sim g$ and $g \sim h$ implies $f \sim h$) is a consequence of the fact that the union of

two sets of measure 0 has measure 0.

Proposition 1.6.3

If $f \sim g$, then, for every $E \in \mathfrak{M}$,

$$\int_E f d\mu = \int_E g d\mu.$$

Theorem 1.6.4

Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $\mathfrak{M}^*$ be the collection of all $E \subset X$ for which there exist sets A and $B \in \mathfrak{M}$ such that $A \subset E \subset B$ and $\mu(B - A) = 0$, and define $\mu(E) = \mu(A)$ in this situation. Then $\mathfrak{M}^*$ is a σ-algebra, and μ is a measure on $\mathfrak{M}^*$.

Remark.
This extended measure μ is called ***complete***, since all subsets of sets of measure 0 are now measurable; the σ-algebra $\mathfrak{M}^*$ is called the ***μ-completion*** of $\mathfrak{M}$. The theorem says that every measure can be completed, so, whenever it is convenient, we may assume that any given measure is complete; this just gives us more measurable sets, hence more measurable functions.